

Convergence of flow-based generative models via proximal gradient descent in Wasserstein space

Xiuyuan Cheng¹, Jianfeng Lu^{1,2}, Yixin Tan¹, and Yao Xie³

¹Department of Mathematics, Duke University

²Department of Physics and Department of Chemistry, Duke University

³H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

Abstract

Flow-based generative models enjoy certain advantages in computing the data generation and the likelihood, and have recently shown competitive empirical performance. Compared to the accumulating theoretical studies on related score-based diffusion models, analysis of flow-based models, which are deterministic in both forward (data-to-noise) and reverse (noise-to-data) directions, remain sparse. In this paper, we provide a theoretical guarantee of generating data distribution by a progressive flow model, the so-called JKO flow model, which implements the Jordan-Kinderlehrer-Otto (JKO) scheme in a normalizing flow network. Leveraging the exponential convergence of the proximal gradient descent (GD) in Wasserstein space, we prove the Kullback-Leibler (KL) guarantee of data generation by a JKO flow model to be $O(\varepsilon^2)$ when using $N \lesssim \log(1/\varepsilon)$ many JKO steps (N Residual Blocks in the flow) where ε is the error in the per-step first-order condition. The assumption on data density is merely a finite second moment, and the theory extends to data distributions without density and when there are inversion errors in the reverse process where we obtain $\text{KL}-\mathcal{W}_2$ mixed error guarantees. The non-asymptotic convergence rate of the JKO-type $\mathcal{W}_2$ -proximal GD is proved for a general class of convex objective functionals that includes the KL divergence as a special case, which can be of independent interest. The analysis framework can extend to other first-order Wasserstein optimization schemes applied to flow-based generative models.

1 Introduction

Generative models, from generative adversarial networks (GAN) [28, 30, 37] and variational auto-encoder (VAE) [41, 42] to normalizing flow [44], have achieved many successes in applications and have become a central topic in deep learning. More recently, diffusion models [33, 66, 67] and closely related flow-based models [2, 3, 26, 50, 74] have drawn much research attention, given their state-of-the-art performance in image generations. Compared to score-based diffusion models, which are designed for sampling, flow models have certain advantages due to their direct capability in estimating likelihood, a basis for statistical inference. However, despite the empirical successes, the theoretical understanding and guarantees for flow-based generative models remain limited.

In this paper, we provide a theoretical guarantee of generating data distribution by a “progressive” flow model, mainly following the JKO flow model in [74] but similar models have been proposed in, e.g., [4, 54, 70]. We prove the exponential convergence rate of such flow models in both (data-to-noise

*The authors are listed alphabetically.

and noise-to-data) directions. Below, we give an overview of the main results. We provide a brief introduction of the most related types of flow-based models, particularly the progressive one, in Section 1.1. A more complete literature survey can be found in Section 1.2.

An abundance of theoretical works has provided the generation guarantee of score-based diffusion models [11, 14, 17, 20, 47, 48, 59]. In comparison, the theoretical study of flow-based generative models is much less developed. Most recent works on the topic focused on the generation guarantee of the Ordinary Differential Equation (ODE) reverse process (deterministic sampler) once a score-based model is trained from the *forward* Stochastic Differential Equation (SDE) diffusion process [16, 18, 49]. For generative models which are flow-based in the forward process, generation guarantee for flow-matching models under continuous-time formulation was shown in [3] under $\mathcal{W}_2$, and in [2] under the Kullback–Leibler (KL) divergence by incorporating additional SDE diffusion. The current paper focuses on obtaining the theoretical guarantee of the JKO flow model [74], which is progressively trained over the Residual Blocks (steps) and generates a discrete-time flow in both forward and reverse directions. The mathematical formulation of the JKO flow is summarized in Section 3, where we introduce needed theoretical assumptions on the learning procedure.

Our analysis is based on first proving the convergence of the forward process (the JKO scheme by flow network), which can be viewed as an approximate proximal Gradient Decent (GD) in the Wasserstein-2 space to minimize $G(\rho)$, a functional on the space of probability distributions. While the convergence analyses of Wasserstein GD and proximal GD have appeared previously in literature [40, 63], our setup differs in several ways, primarily in that we consider the JKO scheme, which is a “fully-backward” discrete-time GD. For the N step discrete-time proximal GD, which produces a sequence of transported distributions p_n , we prove the convergence of both $\mathcal{W}_2(p_n, q)$ and the objective gap $G(p_n) - G(q)$ at an exponential rate, where q is the global minimum of G (Theorem 4.3). The convergence applies to a general class of (strongly) convex G that includes the KL divergence $\text{KL}(p||q)$ as a special case. This result echos the classical proximal GD convergence in vector space where one expects an exponential convergence rate for strongly convex minimizing objectives. While exponential convergence is a natural result from the point of view of gradient flow, this convergence result of JKO-type $\mathcal{W}_2$ -proximal GD did not appear in previous literature to the authors’ best knowledge and can be of independent interest.

After obtaining a small $G(p_n) = \text{KL}(p_n||q)$ from the convergence of the forward process, we directly obtain the KL guarantee of the generated density from the data density by the invertibility of the flow and the data processing inequality, and this implies the total variation (TV) guarantee (Corollary 5.2). The requirement for data distribution is to have a finite second moment and a density (with respect to the Lebesgue measure). The TV and KL guarantees are of $O(\varepsilon)$ and $O(\varepsilon^2)$, respectively, where ε is the bound for the magnitude of the Wasserstein (sub-)gradient of the loss function (hence error in the first order condition) in each of the N JKO steps (Assumption 1), and the process achieves the error bound in $N \lesssim \log(1/\varepsilon)$ many steps (each step is a Residual Block).

To handle the situation when the data distribution only has a finite second moment but no density, we apply a short-time initial diffusion and start the forward process from the smoothed density. This short-time diffusion was adopted in practice and prior theoretical works. We then obtain KL and TV guarantee to generate the smoothed density, which can be made arbitrarily close to the data distribution in $\mathcal{W}_2$ when the initial diffusion time duration tends to zero (Corollary 5.3). The above results are obtained when the reverse process is computed exactly with no inversion error. Our analysis can also extend to the case of small inversion error by proving a $\mathcal{W}_2$ -guarantee between the generated density from the exact reverse process and that from the actual computed one. Theoretically, the $\mathcal{W}_2$ -error can be made $O(\varepsilon)$ or smaller assuming that the inversion error can be made $O(\varepsilon^\alpha)$ for some exponent α (Corollary 5.5).

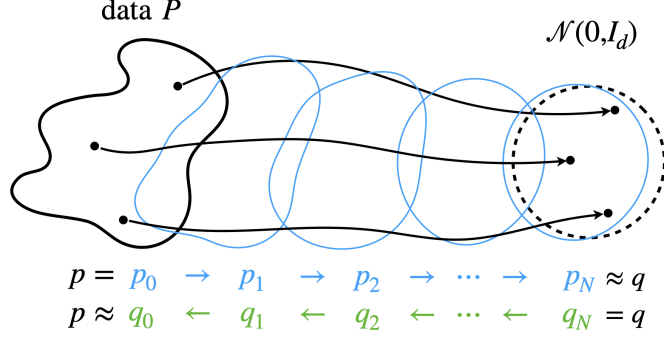


Figure 1: The arrows indicate the forward-time flow from data distribution P to normal distribution q . The forward and reverse processes (19) consist of the sequence of transported densities at discrete time stamps.

1.1 Normalizing Flow models

Normalizing flow. Normalizing flow is a class of deep generative models for efficient sampling and density estimation. Compared to diffusion models, Continuous Normalizing Flow (CNF) models [44] appear earlier in the generative model literature. Largely speaking, CNF models fall into two categories: discrete-time and continuous-time. The discrete-time CNF models adopt the structure of a Residual Network (ResNet) [32] and typically consist of a sequence of mappings:

$$x_l = x_{l-1} + f_l(x_{l-1}), \quad l = 1, \dots, L, \quad (1)$$

where f_l is the neural network mapping parameterized by the l -th “Residual Block”, and x_l is the output of the l -th block. Continuous-time CNFs are implemented under the neural ODE framework [15], where the neural network features $x(t)$ is computed by integrating an ODE

$$\dot{x}(t) = v(x(t), t), \quad t \in [0, T], \quad (2)$$

and $v_t(x) = v(x, t)$ is parametrized by a neural ODE network. The discrete-time CNF (1) can be viewed as computing the numerical integration of the neural ODE (2) on a sequence of time stamps via the forward Euler scheme.

In both categories, a CNF model computes a deterministic transport from the data distribution towards a target distribution q typically normal, $q = \mathcal{N}(0, I_d)$, per the name “normalizing.” The forward time flow is illustrated in Figure 1. Taking the continuous-time formulation (2), let P be the data distribution with density p , $x(0) \sim p$, and denote by $p_t(x) = p(x, t)$ the probability density of $x(t)$. Then p_t solve the continuity equation (CE)

$$\partial_t p_t + \nabla \cdot (p_t v_t) = 0, \quad (3)$$

from $p_0 = p$. If the algorithm can find a v_t such that p_T at some time T is close to q , then one would expect the reverse-time flow from $t = T$ to $t = 0$ to transport from q to a distribution close to p . Note that in the continuous-time flow, invertibility is presumed since the neural ODE can be integrated in two directions of time alike. For discrete-time flow (1), invertibility needs to be ensured either by special designs of the neural network layer type [24, 25, 43], or by regularization techniques such as spectral normalization [9] or transport cost regularization [58, 72].

A notable advantage of the flow model is the computation of the likelihood. For discrete-time flow (1), this involves the computation of the log-determinant of the Jacobian of f_l . For continuous-time